3(a)	Separation: No change, since λ, D and a are constants. Contrast: decreases, since polarization causes intensity to decrease, hence bright fringe is less bright, dark fringe is just as dark.
(b) (i)	The waves of equal f and amplitude in opposite direction interfere/superpose producing a stationary wave. Stationary wave has nodes and antinodes distance between a node and anti-node is $\lambda/4 = 0.75$ cm
(ii)	Resultant wave is the vector sum of the individual waves. Amplitude of the resultant wave = $A\sqrt{2}$. Intensity is proportional to square of amplitude, hence signal intensity is halved.
(c) (i)	All wavelengths meet constructively at that point
(ii)	$\frac{1 \times 10^{-3}}{200} \sin\theta = 1\,\lambda$ $= 700 \times 10^{-9}$ and 400×10^{-9} $\theta = 8.05°$ and $4.59°$ distance from X = 3.00 tan 4.59° and 3.00 tan 8.05° = 0.241 m to 0.424 m
(iii)	If the shortest wavelength (violet) of 3rd order diffract at smaller angle than the longer wavelength (red) of the 2nd order, they will interfere. $\frac{1 \times 10^{-3}}{200} \sin\theta = 3 \times 400 \times 10^{-9} \rightarrow \theta = 13.9°$ $\frac{1 \times 10^{-3}}{200} \sin\theta = 2 \times 700 \times 10^{-9} \rightarrow \theta = 16.3°$ Hence, they will overlap.

4 **(a)** gradient = E